

THEMATIC RESEARCH · CYBERSECURITY

The Failure Ratchet

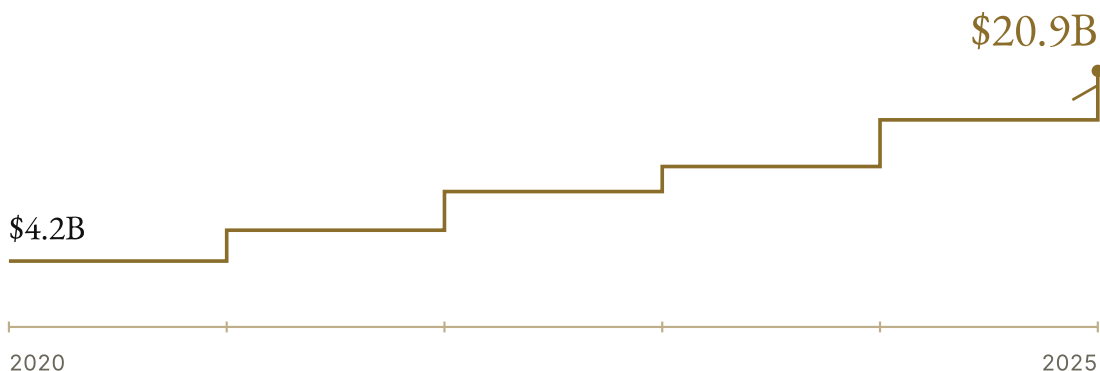
Everyone gets breached, everyone gets paid: cybersecurity as an industry — and what the widest valuation spread in software is really pricing

TON618 Capital Research · Sector thematic · DRAFT note (distilled 2026-08-28 from the archived exhaustive draft).

Gate status as of 2026-08-29: steelman gate PASS (4/5 on re-run); claim-verifier PUBLISHABLE (round-3 re-verification passed 2026-08-29 after two correction rounds). No publish authorized.

Pricing as of 2026-08-28 unless dated otherwise. A separate TON618 workstream covers the Palo Alto Networks valuation; this note is the industry, not any one name.

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**\$4.2B → \$20.9B**

reported US cybercrime losses, 2020 → 2025 (FBI). The product category's failure metric, up fivefold in five years.

~8.5 million

Windows machines disabled by CrowdStrike's own faulty update in July 2024 (Microsoft's estimate). CrowdStrike's recurring revenue is up 51% since.

~\$85 billion

three security mega-deals closed inside twenty-four months: Splunk into Cisco, Wiz into Google, CyberArk into Palo Alto Networks. The last closed March 11, 2026.

46x vs 13.6x

price-to-sales on CrowdStrike versus trailing price-to-earnings on Check Point: one industry, priced as three different asset classes.

>\$20 billion

Microsoft's security revenue at last disclosure (January 2023 — it has not updated the figure since). Roughly what the three largest pure-plays bill combined *today*, on a figure three and a half years stale.

Before the verdict — the constructive case on price, in its own terms. This note ends up skeptical of what the market pays for most of this group, so the side owed the strongest opening statement is the one that thinks the prices are right: this is a mandated-demand market where the frontier compounds revenue at 21–38% year-over-year on gross margins that run to the low 80s (group median 77%), several of the marquee names carry net cash, and the buyer of last resort is a strategic acquirer that just paid 26% premiums for scarcity — three of the last four scale assets cleared at prices the trailing-multiple crowd called impossible. Trailing price-to-sales is the wrong lens for that case: it ignores net cash, ignores forward growth, and ignores that recurring revenue under multi-year contracts is a harder asset than the sales of almost any other industry at the same multiple. What this note tests of that case: the growth (on one clean basis, \$5), the deal record (\$4), and the durability mechanism (\$2). What it cannot test within its frame: terminal margins and per-name discounted value — those need the single-name valuation engines, and we say so where it matters.

The verdict

The question this note answers is the one anyone who reads both the breach headlines and the stock charts eventually asks: **if the attackers keep winning, why do the defenders' stocks keep winning too — and can both lines keep going up?**

The answer, in one mechanism: cybersecurity is the only industry we cover where the measurable failure of the product category *grows* the market. Nobody buys more of a drug that doesn't work. But a breach headline is unpaid marketing for the breach-prevention industry, because the buyer is not purchasing outcomes — the buyer is purchasing *defensibility*: against regulators (a lengthening calendar of mandatory-spend rules), against boards and insurers (fear with a budget line), against the plaintiff's bar. Fear and compliance both ratchet in one direction. We call it **the failure ratchet**, and it is real: losses at records, disclosed software vulnerabilities up 2.8x in eight years, and a rulebook — SEC four-day incident disclosure, Europe's NIS2 and DORA, the Pentagon's contractor certification regime — that converts each of those records into next year's mandatory budget.

But the ratchet turns for the *industry*, not for every vendor in it — and that distinction is the whole investment question. Enterprise security buyers spent the last three years consolidating vendors, and the sellers responded with the largest acquisition wave in the sector's history. The result sits in our comp spread as the widest valuation dispersion of any software vertical we track: from 1.1x sales to 46x sales inside one fifteen-name universe — a momentum asset, a value stock, and everything between, all nominally in the same business. That spread is not confusion. It is the market pricing *survivorship in the consolidation*: pay almost anything for the three or four presumed platform winners, pay cash-flow multiples for everything presumed to be consolidated, commoditized, or bled.

Our verdict on the group: the demand is real and substantially mandated — this is not a fear bubble — but at current prices most of the universe is already paid as if it individually owns a mechanism that only the platform winners will. The honest output is a sorted watchlist, not a sector call, and the sort itself — who survives the consolidation, and whether the industry's first genuine deflation forces (Microsoft's bundle, AI agents in the security operations center) shrink the pie or merely re-slice it — is the argument of this note.

The ratchet, measured

The demand driver of this industry is the threat environment, so before any stock: the threat tape, from primary sources.

WHAT	LATEST	TRAJECTORY
Reported US cybercrime losses (FBI complaint center)	\$20.9B in 2025 ; first year above 1M complaints	\$4.2B (2020) → \$10.3B (2022) → \$16.6B (2024) → \$20.9B — 5x in five years
Software vulnerabilities disclosed per year (CVEs — the catalog numbers assigned to publicly known flaws)	49,972 in 2025 ; 2026 through Aug 28 already 59,236 (partly backlog processing)	18,113 (2017) → 26,431 (2022) → 49,972 (2025) — up 2.8x in eight years (our basis: NVD publication date, later-rejected records included; other counting conventions yield different levels)
Vulnerabilities confirmed exploited in the wild (CISA's Known Exploited Vulnerabilities catalog)	1,685 cataloged	187 added in 2023, 186 in 2024, 245 in 2025; 201 so far in 2026 — on pace (~300 annualized) to exceed every year since the catalog's 2021-22 launch backfill
Global security spending (Gartner forecast — an analyst estimate, not a measurement)	~\$213B in 2025; 2026 forecast raised to ~\$248.9B in Gartner's mid-2026 update (from ~\$240B in the July 2025 vintage)	+12-13%/year

Password attacks blocked by
Microsoft's own systems

>7,000 per second (2024 report);
identity attacks +32% in H1 2025
(2025 report)

up and to the right

Two honesty notes that survive from the full draft. The vulnerability counts partly reflect the cataloging agency working through a processing backlog — the decade trend is unambiguous, single-year jumps are not. And "reported losses" means reported: the series undercounts, and some of its growth is reporting propensity. We use these series exactly the way the industry's buyers do — as the fear index — because that is the point: the fear index only rises, and the compliance calendar (§7) has been legislated to follow it.

One series on the demand side has actually turned down: the US federal government's own cyber budget. CISA — the civilian cyber-defense agency — was cut roughly 17% in the administration's FY2026 request, which took positions down 29% (3,292 → 2,324); the FY2027 request asks for a further net reduction of roughly \$360-386M (gross program cuts of \$707M, partly offset by transfers) and lands headcount at 2,865. The buyer whose threat warnings legitimize everyone else's budget is trimming its own. That asymmetry — private mandated demand ratcheting while public discretionary demand is cut — runs through everything below.



The stack, and how the money is made

Enterprise security is a stack of control points, each historically its own market:

- **Network security** → **SASE** (Secure Access Service Edge — delivering the corporate firewall and "zero-trust" access, where every user authenticates to each application instead of getting network-wide VPN access, as a cloud service): Palo Alto Networks, Fortinet, Check Point, Zscaler, Cloudflare.
- **Endpoint** → **XDR** (software agents on laptops and servers detecting malicious behavior; XDR extends the detection across email, cloud, and identity): CrowdStrike, SentinelOne, Microsoft Defender.
- **Identity** (who may log into what — including privileged-access management, the vaulting of administrator credentials): Okta, SailPoint, Microsoft Entra, and CyberArk — now inside Palo Alto.
- **Cloud security** (scanning cloud infrastructure and code for exploitable exposure): Wiz — now inside Google — plus the platforms' modules.
- **Security operations** (the SOC — security operations center — and its data layer, the SIEM, which aggregates logs and detections): Splunk — now inside Cisco — plus Microsoft Sentinel and the platforms.
- **Vulnerability management, data security, resilience**: Qualys, Tenable, Rapid7; Varonis; Rubrik.
- **Consumer**: Gen Digital (Norton, LifeLock, Avast).

Note how many layer leaders that list places *inside someone else* as of this year. That is \$4.

The business model across the stack is subscription land-and-expand, measured in **ARR** (annual recurring revenue — the annualized value of active subscriptions): sell one module, then expand seats and cross-sell until existing customers grow spend every year. The economics it produces, from our comp spread (14 engine names plus a hand-built Check Point row; as-reported SEC filings; engine details in the appendix): gross margins of 64-83%, median 77% — genuine software economics. And yet **the median pure-play still loses money under GAAP** (median operating margin -2.9%), because stock-based compensation — employee pay issued as shares, a real cost that vendors' "non-GAAP" margins add back — runs a fifth of revenue at the median. Free cash flow is strong (26% median margin) but structurally flattered twice: subscriptions collect cash up front, and stock comp never touches the cash-flow statement. The symmetrical caveat: the GAAP losses those cash flows sit beside are themselves depressed at the acquisitive names by purchase-accounting amortization and deal costs — Palo Alto's latest quarter swung to a GAAP loss on exactly that. We quote GAAP first, always, and haircut both directions.

The exceptions prove the structure. Fortinet designs its own custom chips (ASICs) for traffic inspection, giving it a price-performance moat in physical appliances and industrial-edge security — and it converts that into the sector's best real profitability: a 34% GAAP operating margin *while growing 26%* (Q2 2026). Check Point, the original firewall vendor, runs the same physics a decade further on: an enormous installed base, ~1% growth, a net-cash balance

sheet (symmetry note: its latest-quarter cash includes \$1.8B raised in a \$2B convertible offering — the same instrument we hold against the mid-caps, so it is counted here too), and a 13.6x trailing earnings multiple — the sector's value stock, and the live experiment in what an ex-growth security franchise is worth.

And the demand mechanism, stated fairly. The naive version: security spend is fear- and compliance-driven, not return-on-investment-driven — no CFO can compute the return on the breach that didn't happen, but every CFO can compute the cost of being the company that skipped the control the regulator asked about. The fairer version, which the last three years demand: buyers *do* run ROI math — it is just consolidation ROI. After 2022, chief information security officers got budget scrutiny and started counting vendors and consoles. Both versions produce the same industry-level ratchet; they produce opposite per-vendor outcomes. The comp spread says the market believes the second version. So do we.



The gravity well and the platform wars

Microsoft first, because the largest security company on earth is not in the comp spread. In January 2023 Microsoft disclosed that its security business had passed **\$20 billion in revenue** — a figure it has not updated publicly since, which is itself worth pondering. On our own spread, Palo Alto Networks, CrowdStrike, and Fortinet bill about \$20.8B combined on their latest fiscal years: the three largest pure-plays *together* have only just caught up to where Microsoft said it was three and a half years ago. The delivery mechanism is the **E5 bundle**: the top Microsoft 365 enterprise license includes serviceable versions of endpoint protection, email security, identity, and compliance tooling — four or five entire pure-play categories — at a marginal price no standalone vendor can meet. This is the structural case against every pure-play in the universe, and it is strongest where security is closest to an IT feature (workforce identity, email) and weakest where security is adversarial, physical, or must stay neutral across clouds — a buyer securing Amazon and Google workloads with Microsoft-only tooling is handing the auditor a finding.

Against that gravity well, the pure-plays' answer is **platformization** — become one of the few consolidated platforms a vendor-cutting customer keeps — and the race produced the largest deal wave in the sector's history — three completions inside twenty-four months (March 2024 to March 2026), all verified against the acquirers' filings and releases:

DEAL	ANNOUNCED	CLOSED	PRICE	WHAT WAS CONSOLIDATED
Cisco – Splunk	Sep 2023	Mar 2024	~\$28B	the SOC's data layer, into a networking giant
Google – Wiz	Mar 2025	Mar 2026 (cleared unconditionally)	~\$32B	cloud security's leader, into a hyperscaler
Palo Alto – CyberArk	Jul 2025	Feb 11, 2026	~\$25B (\$45.00 cash + 2.2005 PANW shares per share; 26% premium)	privileged identity, into the platform pure-play

Second-tier consolidation ran the same direction: Darktrace to private equity (~\$5.3B, 2024), Recorded Future to Mastercard (\$2.65B, 2024), Secureworks to Sophos (2025), three Check Point tuck-ins (2026). The pattern, not the deals: **platforms buying category leaders, at premiums, to shorten the race** — and the shelf of independent scale assets is nearly bare (\$7.4).

Is platformization winning? Both sides have real numbers. Palo Alto's next-generation security ARR reached \$8.1B, +60% — but that includes \$1.6B *acquired*, so the organic rate is materially lower. CrowdStrike's consolidation vehicle (Falcon Flex, a flexible spend commitment across its modules) more than doubled year-over-year to \$2.29B of its \$5.84B ARR. Meanwhile the best single-category products still out-grow every platform organically — Cloudflare +36% revenue and accelerating, Rubrik +38% revenue, Zscaler +21% ARR excluding its own acquisition (its only disclosed ex-acquisition figure; organic revenue is not broken out) — because in security, unlike office software, a second-best control is a liability you can measure in incident response. The honest synthesis: platforms are winning *procurement*; best-of-breed keeps winning *product*; which is why the platforms keep having to buy the best-of-breeds. That loop is the industry's operating system.

The scoreboard

(Prices and market caps 2026-08-28. One basis rule, both columns: every growth figure AND every GAAP operating margin is the company's LATEST REPORTED QUARTER, year-over-year, as of 2026-08-29 — no mixing of annual and quarterly bases (refresh script archived with the data); the single exception is SailPoint's margin, fiscal-year-tagged because no quarterly GAAP figure is in our receipts. P/S = market cap ÷ latest fiscal-year revenue — plain price-to-sales because enterprise-value versions were not computable cleanly for the whole group; the appendix explains. And say the caps out loud: P/S ignores the balance sheet in both directions — it penalizes the net-cash names here as surely as it flatters the levered ones. Two further caps, stated: these are trailing multiples on latest fiscal-year revenue, which mechanically penalize the fastest growers — Palo Alto is the only row re-based to a forward guide because it is the only guide inside our receipts — and no row credits net cash. Both re-cuts land with the September filings refresh, and both move in the constructive case's favor. Full tables, per-name detail, and the short-interest annex live in the archived draft.)

	NAME	PX / MCAP	P/S	GROWTH (LATEST QTR)	GAAP OP MARGIN	THE ONE-LINE READ
PLATFORMS	Palo Alto (PANW)	\$372.18 / \$303B	32.9x (≈27x on FY26 guide)	+31% (acq.-assisted)	GAAP loss (q; deal costs)	the platform thesis incarnate, mid-digestion of its biggest deal ever; FY27 guide lands Sept 1
	CrowdStrike (CRWD)	\$217.60 / \$222B	46.0x	+26%, ARR \$5.84B	-2.3% (q)	best moat exhibit in the sector (\$5.1), richest multiple in the sector
PROFIT ENGINES	Fortinet (FTNT)	\$165.78 / \$122B	17.9x	+26%	+34% (q)	custom-silicon economics; the group's only fat-margin grower
	Qualys (QLYS)	\$187.05 / \$6.5B	9.7x	+11%	+34% (q)	small, profitable, heavily shorted (11.7% of shares)
	Check Point (CHKP)	\$138.15 / ~\$14B	5.2x · 13.6x P/E	+~1%	+27% (q; FY25 was 30.5%)	the value experiment: installed base vs. melt rate
EDGES & ARCHITECTS	Cloudflare (NET)	\$301.61 / \$97B	44.9x	+36%, accelerating	-29.5% (q; AI investment)	fastest grower; a network that added security — kept with that caveat
	Zscaler (ZS)	\$184.45 / \$30B	11.2x	+25%; ARR \$3.5B +25% (+21% ex-acquisition — an ARR figure; no organic revenue disclosed)	-3.5% (q)	architecturally right, commercially squeezed; a quarter of Cloudflare's multiple on comparable growth
IDENTITY	Okta (OKTA)	\$166.20 / \$28B	9.5x	+11%	+13% (q)	the last neutral identity platform of scale — the note's most interesting tension (\$8)
	SailPoint (SAIL)	\$19.84 / \$11B	10.5x	+21.6%	-29% (fy — no quarterly GAAP figure in receipts)	re-IPO'd governance player; private-equity overhang is a for-sale sign on a timer
THE SHELF	SentinelOne (S)	\$21.49 / \$7B	7.2x	+21%, ARR \$1.22B	-31% (q)	last independent next-gen endpoint asset; perpetual strategic chatter, unconfirmed
	Rubrik (RBRK)	\$92.95 / \$15B	11.4x	+38%; sub ARR +33%	-16.8% (q)	the growth asset of the mid-caps; convert-heavy balance sheet
	Varonis (VRNS)	\$43.92 / \$5B	8.1x	+18%; SaaS ARR +52% (+25% ex-conversions)	-22.6% (q)	mid-transition to SaaS; transition optics
	Tenable (TENB)	\$37.70 / \$4.2B	4.2x	+8.6%	+4.6% (q; FY25 was -0.9% — the sign flips on the quarterly basis)	cheap with balance-sheet reasons (\$5.2) — and GAAP-profitable in its latest quarter
CONSUMER	Rapid7 (RPD)	\$13.45 / \$0.9B	1.1x	-1.5%; ARR -2%; 12% headcount cut (Aug 2026)	+1.4% (q)	the broken one — now shrinking; highest short interest in the group (11.8%, 98th percentile)
	Gen Digital (GEN)	\$30.79 / \$18B	3.7x · 8.3% FCF yield	+6% GAAP (+11% non-GAAP incl. acquisition stub)	+33.2% (q, falling)	cheapest large name — and our forensic pass says the yield is compensation, not opportunity (\$5.2)

The headline is the spread itself. From 1.1x to 46x sales is a forty-two-fold gap inside one industry — wider than anything else we run this engine on. On the clean one-quarter basis, the growth gap reads: **mature half -1.5% to +18%, frontier +21% to +38%, full-universe median +21%** (disclosure: "frontier" is defined as $\geq 20\%$ latest-quarter growth, so part of the 18-vs-21 adjacency is the cut-point itself — the -1.5%-to-+38% span is not). The same table holds a momentum asset, a 13.6x-earnings value stock, an 8%-yield consumer name, and a shelf of loss-making mid-caps. Growth-adjusting narrows but does not close it — and produces one twist worth printing: per point of latest-quarter growth, the most expensive stock in the group is not CrowdStrike (~1.8x P/S per point) but Check Point (~5x per point on ~1% growth) — the "value stock" is only value if the melt stops. The market is not pricing "cybersecurity"; it is pricing survivorship — and roughly along the lines \$4 predicts: premiums for the consolidators and the Microsoft-orthogonal (elite endpoint, custom silicon, consolidated contracts), discounts for the Microsoft-adjacent and the consolidatable.

📌 5.1 The outage that proved the moat

On July 19, 2024, a faulty CrowdStrike content update — twenty-one input fields where the software expected twenty, per the company's own root-cause analysis — crashed roughly 8.5 million Windows machines worldwide, grounding airlines and stopping hospital systems; by most accounts the largest IT outage in history. The disclosed direct financial impact: about \$60 million of near-term new business — about 1.6% of the company's ARR at the time. Two years later, ARR is **51% higher** (\$3.86B at the outage, \$5.84B today) and the company just called its latest quarter the best in its history. Read it precisely: the category's most public failure in a decade did not durably hurt *the company that caused it*. That is the failure ratchet operating at maximum: switching costs dominate measured performance, and the industry is structurally insulated from its own outcomes. It is the best evidence for the platforms' moats — and, as \$9 argues, the most unsettling fact in the sector.

+51%

CrowdStrike's annual recurring revenue growth in the two years after its own faulty update disabled ~8.5 million Windows machines. The category's most public failure in a decade did not durably hurt the company that caused it.

🔍 5.2 Cheap, with reasons

House rule: the cheapest names get an accounting check before anyone calls them mispriced. We ran the full forensic engine on the three cheapest. **Gen Digital: SERIOUS RED FLAGS** — a distress-zone solvency score, a receivables spike, a goodwill-heavy balance sheet three consumer acquisitions deep, ~3x net debt to EBITDA. Not proof of misstatement; proof that the 8.3% free-cash-flow yield is priced compensation for a levered roll-up whose organic economics are hard to see. **Tenable and Rapid7: ELEVATED CONCERN**, on matching patterns — goodwill, leverage, convertible-note stacks — which also caps what any acquirer would pay. The value end of this sector is not a refuge; it is a different risk.



Where we are in the cycle

Mid-cycle for the industry; late-cycle for the point-product business model. The threat tape (\$2) is at records, aggregate spend estimates still compound double-digit — and yet, on the one-quarter basis, the mature half of the universe grows -1.5% to +18% while the frontier grows +21% to +38%. That is not a demand cycle rolling over; it is share moving from point products to platforms and edges, exactly as the procurement story predicts. In that regime the sector beta stops meaning much: the *industry* metric compounds while half the universe ex-growths, and dispersion — not direction — becomes the trade. The one genuinely cyclical vertical is US federal (\$2's cuts; concentrated pain in federal-heavy franchises like Tenable's), and it has turned down for the first time in the sector's modern history while

Brussels escalates enforcement in the opposite direction. Two sovereigns, two directions — a regional-mix story before it is a sector story.



CLICK 7

The catalysts

7.1 AI, on both sides of the wall

The attack side is a demand tailwind with primary - source markers: Microsoft's own telemetry blocks over 7,000 password attacks per second (its 2024 report), identity attacks rose 32% in the first half of 2025 (its 2025 report), and the FBI's record \$20.9B loss year was led by the two fraud categories AI most obviously industrializes. Attribution of the increase to AI specifically is mostly vendor marketing; the volume records and the mandatory -disclosure rules that turn attacks into budget events are not.

The defense side cuts the other way, and deserves respect: every vendor now ships an "agentic SOC" — AI agents doing the triage work of human security analysts (Microsoft's Security Copilot agents; CrowdStrike's Charlotte AI, claiming forty analyst -hours saved per week — vendor claims, labeled). Security operations is where security spending is substantially *labor*, and agents compress labor. If a tier -one analyst's queue can be triaged by software bundled into a platform contract, the managed -services hours and per -seat tooling that analyst used are the first casualties. This industry has commoditized every product layer it ever invented — antivirus, firewalls, log pricing — and always escaped by inventing the next layer up. AI is the first force that commoditizes the operations layer too. Whether the next layer is bigger than the one being compressed is the sector's genuine open question, and we arm a tripwire on it rather than resolve it by assertion (§10).

The strongest version of "the next layer": **non -human identity**. AI agents hold credentials — service accounts, API keys, certificates — and the industry's own survey data (from CyberArk, now published under Palo Alto's banner; a vendor survey, labeled as such) counts **109 machine identities per human identity**, up from 82 a year earlier. Every deployed agent is a privileged credential that never sleeps and can be prompt -injected. That is the stated logic of the sector's biggest -ever deal — \$25B for the privileged -access leader — and the tell is dated: if agent identity is a real market and not a slideware category, it shows up first in Palo Alto's identity attach (guide: September 1) and in whether Okta's and SailPoint's growth re -accelerates.

7.2 The quantum migration — the mandated catalyst

(Cross-linked to the house quantum series — v1.0, June 2026, and the August 2026 re-score — and consistent with its clocks and tripwires. Nothing here asserts Q-Day is near.)

A cryptographically relevant quantum computer would break the public -key encryption underlying secure communication. The house series puts code -breaking -relevant capability at roughly **35 -40% odds within ten years**, with commercial fault tolerance a 2030s story; the current laboratory frontier is ~70 claimed logical qubits against the ~1,200 the series benchmarks as needed to threaten a 256 -bit elliptic -curve key. This note moves none of those numbers.

The investable point is different: **the defensive migration is mandated regardless of whether the machine ever arrives**. The driver is "harvest now, decrypt later" — adversaries recording encrypted traffic today to decrypt the day a capable machine exists — which makes long -secrecy data *already* exposed, and which governments have therefore legislated against on fixed calendars. The post -quantum cryptography (**PQC**) standards are finished: NIST finalized the ML -KEM encryption and ML -DSA signature standards in August 2024. The NSA's transition schedule for national -security systems gates *procurement* from January 2027 and requires full transition by 2030 -2033 depending on system type (dates from NSA documents that resisted automated verification — flagged for the pre -publish pass). And in June 2026 the federal civilian calendar hardened at its sensitive core: the **June 22, 2026 White House executive order on post -quantum readiness** (implemented through OMB memorandum M -26 -15) sets dated milestones — quantum -resistant *encryption* by **December 31, 2030** and quantum -resistant *authentication* by **December 31, 2031** — scoped to high -value assets and high -impact/highly -sensitive systems, "as feasible," with full migration retained as the final phase by **2035**. Narrower than a government -wide deadline, but it converts the most sensitive slice of the 2035 goal into dated, phased procurement obligations. The federal civilian migration alone carries a ~\$7.1B official cost estimate for 2025 -2035, a floor that excludes defense and intelligence. And deployment

is already measurable: by Cloudflare's traffic data, post-quantum-encrypted web traffic went from 29% of human browsing at the start of 2025 to about 52% by December, inflected by Apple's OS releases.

Who gets paid is the part the market persistently gets wrong. Not, on this evidence, the quantum-computing stocks (the house series covers that trade). The migration is a decade of cryptographic inventory - and - replacement inside firewalls, VPNs, certificates, and key management — revenue that accrues to **crypto-agile incumbents** as a mandated refresh cycle (the network platforms and appliance vendors of \$5, whose gear carries the cryptography) and to the certificate-lifecycle specialists, most of which are private or already acquired. No public pure-play in our universe discloses material PQC revenue today; any note quoting one would be manufacturing precision. The catalyst is the calendar: it converts a physics debate into a compliance budget — and if the quantum series' tripwires ever fire, that budget goes from scheduled to panicked.

7.3 The regulation calendar

Every item verified against the regulator's own text (details in the appendix). Binding now: the SEC's four-business-day incident-disclosure rule (in force since December 2023 — with a live industry petition to rescind it, worth watching under a deregulatory Commission); the EU's **DORA** (operational-resilience rules for finance, applied January 2025) and **NIS2** (the largest expansion of mandatory security baselines in EU history — transposition so late that the Commission referred four member states to the EU Court of Justice in July 2026, which means most of that demand is still *arriving*); and **CMMC**, the Pentagon's certification regime, a condition of contract award since November 2025 phasing across a defense industrial base the rule's own analysis puts at ~220,000 companies, through 2028. Add the June 2026 post-quantum executive order (§7.2): its 2030/2031 milestones for agencies' most sensitive systems are now part of this same mandatory-spend calendar.

Dated, inside ninety days: the EU Cyber Resilience Act's vulnerability-reporting obligations apply **September 11, 2026**; the long-delayed US critical-infrastructure incident-reporting rule (CIRCIA — 72-hour reporting across sixteen sectors) is projected for **September 2026**, having already missed one statutory deadline; and the twice-extended Cybersecurity Information Sharing Act expires again **September 30, 2026**. Against all of that, the same US government mandating private spend is cutting its own cyber agency (§2) — the sector's compliance tailwind and its sentiment headwind currently share a return address.

7.4 Consolidation: the shelf is nearly bare

The deal wave (§4) established that strategic buyers pay premiums for category leaders and that regulators cleared even a hyperscaler buying the cloud-security leader, unconditionally. What remains on the board: in identity, only Okta and SailPoint; in next-gen endpoint, only SentinelOne; in security operations and cloud security, effectively nothing of scale. The mid-cap shelf (Qualys, Tenable, Rapid7, Varonis, Rubrik) is textbook consolidation feedstock — except our forensic pass shows the convert stacks an acquirer would inherit. And the two-sided catalyst nobody prices: Palo Alto's digestion itself. If the CyberArk integration visibly works, the platform-buys-leader playbook gets re-run — by Palo Alto again, by CrowdStrike, possibly by Google. If it visibly struggles, the premium the market pays for "consolidation winners" is exactly the thing that re-rates. This is a scarcity map with dated observables, not a prediction; we assign no take-out odds and imply no positions.



The deeper case against the sector itself

(Two sides get steelmanned in this note, and they are different sides. The block before §1 stated the constructive case for the sector's PRICES — the side this note's valuation verdict cuts against. This block states the skeptic's case against the sector's DEMAND STORY — the side the note's industry verdict cuts against. In its own strongest terms, before we test it:)

Microsoft ends every pure-play's story; it is only a question of when. The largest security vendor is not in the spread. Its last disclosed security revenue — \$20B, three and a half years ago, never updated — is a figure the three biggest pure-plays *combined* have only now caught up to, and it ships inside a license the customer already buys, at a marginal price no standalone vendor can meet. The pure-plays' own strategy — consolidate into platforms — concedes the endgame: three or four bundles,

and Microsoft owns the biggest bundle in enterprise history. You are paying 10-46x sales for companies whose terminal competitor gives the product away.

The "ratchet" is a budget reallocation marketed as a growth market. Aggregate spend grows 12-13% on the vendors' favorite analyst estimate, but the mature half of the universe grows -1.5% to +18% on the latest quarter — one of them is now outright shrinking. Buyers are not writing bigger checks out of fear; they are consolidating vendor count under CFO pressure — a zero-sum reallocation among sellers. The platform headline rates are flattered by acquired revenue and repackaged commitments; strip the wrappers and the industry's organic rate is its customers' budget growth — high single digits — while half the universe is priced for permanent twenty.

AI deflates this industry before it inflates it. Security's revenue pool is substantially labor-adjacent — analyst seats, managed-services hours, per-seat tooling — and the agentic products every vendor proudly launched are labor-compression machines whose marketing (forty analyst-hours saved a week) is deflation math stated as a feature. Every prior product layer here commoditized within a decade; AI is the first force that commoditizes the operations layer too. The constructive case assumes the savings get re-spent inside security. Nothing guarantees that.

The accounting says the sector never earned its multiple. Median GAAP operating margin: negative, at scale, fifteen years into the subscription era. Stock compensation: a fifth of revenue at the median. The cash flow is flattered by up-front collection and the stock-comp add-back. The cheapest large name carries a serious-red-flags forensic verdict; the cheap mid-caps carry convertible stacks; the best real GAAP business trades at 61x trailing earnings.

Even the demand floor has a crack. The fear-buyer of last resort — the US government — cut its own cyber agency, cut its cyber-agency positions 29% in one budget request, let its information-sharing statute lapse in 2025, and now runs it on short-term extensions. And the CrowdStrike outage, read coldly, shows buyers stayed not because the product is irreplaceable but because they *cannot leave*. Industries whose customers are captive and whose failures are consequence-free eventually meet the regulator, or the substitute.



Testing it

On Microsoft: partly conceded — and already priced, name by name. The strong form fails on a decade of evidence: CrowdStrike built \$5.8B of ARR *while* Defender was effectively free in the bundle; Okta still holds workforce identity at scale against bundled Entra; the layers Microsoft's bundle could clear, it cleared years ago. Security is adversarial — "good enough" is a measurable liability after the first incident — and cross-cloud neutrality cannot be bundled away by definition. What survives: the bundle is the correct explanation for the *dispersion*. It is exactly the Microsoft-adjacent names that trade at the discounts and the Microsoft-orthogonal names that hold the premiums. The market has graded this claim name by name, and roughly correctly.

On reallocation-versus-growth: the skeptic's best point, and it mostly survives. Our own spread says growth ex-frontier runs -1.5% to +18% with a median of +8.6% — though the full universe, frontier included, medians +21% on the same one-quarter basis, and fairness requires printing both — and we showed the acquired-ARR flattery ourselves. Where it overreaches: the regulation calendar is not budget-neutral. NIS2, DORA, CMMC, and the CRA create spend obligations for hundreds of thousands of entities that previously spent near zero — Europe's mid-market, the US defense supply chain — arriving on statutory schedules through 2028. The ratchet is real at the bottom of the pyramid even while the top consolidates.

On AI deflation: unresolved, and we refuse to resolve it by assertion. The evidence on both sides is vendor claims; no managed-services price crack or disclosed agent-cannibalization exists in reported numbers yet. It becomes a tripwire (below). We note only that this industry has escaped every prior commoditization by inventing the next control layer, and that non-human identity is the specified candidate — but for the first time, the deflation mechanism is specified too.

On the accounting: conceded as description, contested as verdict. Every figure is ours and stands. For the platform names it is the standard growth-software bargain — and Okta, GAAP-profitable since mid-2025 and expanding that margin to 13% in its latest quarter, shows the model can mature. For the sub-15%-growth shelf carrying 20-30% stock comp and convert leverage, the skeptic simply wins, and the market's 1-8x pricing of that shelf says it agrees.

On the federal crack: conceded as a vertical, rejected as a floor-breaker. It is one customer, single-digit revenue share for most vendors, concentrated where we flagged it — against a private mandated calendar enforced by a different sovereign. The divergence is a regional-mix trade unless the US deregulatory turn reaches the SEC's disclosure rule. That, too, is a tripwire.

Net: the constructive frame holds for the industry; the skeptic wins two partial verdicts that matter for positioning — the top-of-pyramid ratchet is substantially reallocation, and the loss-making mid-cap shelf is structurally impaired rather than cheap. Both verdicts are visible in the dispersion, which is why the dispersion is the note.



What would change our view

The read leans on four assumptions: mandated demand keeps aggregate spend compounding; consolidation transfers the growth to a handful of platforms and edges; the deflation forces sort the universe rather than shrink it; the post-quantum migration is a dated demand floor. Each carries an observable that would flip it. (Drafted, not yet armed — tripwires register in the house scorecard only at publish.)

- **CS-1 — the ratchet breaks:** two consecutive quarters of sub-10% median revenue growth across our spread *while* the loss and exploited-vulnerability series still set records → the fear-to-budget transmission has failed; the core thesis with it. Silence scores as silence.
- **CS-2 — platformization stalls:** Palo Alto's fiscal-2027 guidance (September 1) shows ex-acquisition platform ARR growth below 20%, or the identity attach is de-emphasized on two consecutive calls → the consolidation-winner premium loses its engine.
- **CS-3 — the gravity well accelerates:** Microsoft re-discloses security revenue at \$35B+, or unbundles E5 security into a cheaper standalone tier → erosion becomes displacement; every adjacent discount widens permanently.
- **CS-4 — AI deflation shows up in numbers:** a managed-security provider of scale reports price compression attributed to agentic tooling, or a universe vendor discloses automation-driven seat churn → \$9's open question resolves for the skeptic; sector growth expectations reset down.
- **CS-5 — the US compliance leg folds:** the SEC rescinds its incident-disclosure requirement *and* the critical-infrastructure reporting rule slips past mid-2027 → the regulation calendar flips from tailwind ledger to risk ledger.
- **CS-6 — quantum re-rates (inherited):** the house quantum series' physics tripwires fire → the PQC migration converts from scheduled to emergency demand, and crypto-agile incumbents re-rate. Scored in that series; never re-scored here.
- **CS-7 — prices come to us (the upgrade tripwire):** the frontier de-rates to the low-20s on sales with ARR growth still above 25%, or a platform prints ex-acquisition ARR growth above 25% for two consecutive quarters → the survivorship premium is being paid for a mechanism we can now see working, and the watchlist becomes a buy list. Five of the seven tripwires fire against the sector's pricing; this one fires for it — the reversal is two-directional by construction. One asymmetry worth naming: CS-2 fires on a single guidance print while CS-7's platform leg asks for two consecutive quarters. That is deliberate — a guide is the company's own forward statement, a realized rate has to repeat — and CS-7's price leg fires on one observation.



What is actionable — honestly

Nothing in this note clears the bar of a conviction idea at current prices, and the house treats that as a publishable conclusion. The watchlist the evidence supports:

1. **The identity question (Okta).** The sector's biggest-ever deal was a \$25B statement that identity is the control plane of the AI-agent era. The last independent identity platform of scale trades at 9.5x sales, GAAP-profitable since mid-2025 with the margin still expanding, on 11% growth. If agent identity is real, growth re-accelerates or someone pays the scarcity premium; if the Microsoft bundle grinds on, it is a melting franchise at a fair price. A genuinely contested thesis — which is what makes it the interesting one. Killed by: contracted-backlog growth below ~10%, or CS-3.

2. **The value experiment (Check Point vs Gen Digital).** Check Point at 13.6x earnings with a net-cash balance sheet (part convert-funded — §3's symmetry note) is the clean test of whether ex-growth security holds a cash floor. Gen Digital's 8.3% cash yield is the same test with serious forensic flags and 3x leverage attached — the yield is the risk premium. Treating them as the same "cheap security" trade ignores our own accounting work.
3. **The squeezed architect (Zscaler).** Architecturally right, commercially squeezed, at half Cloudflare's multiple on similar growth. Its fiscal-year guide prints September 3; no view until it does.
4. **Post-quantum exposure.** There is no clean public pure-play, and we decline to manufacture a "quantum-safe basket." The security expression of the mandated migration is incumbent and boring: it is already owned through the network platforms.

What we are deliberately not doing: no take-out speculation (the scarcity map is not a target list), no short on the mid-cap shelf into a live consolidation bid, and no call on the two premium platforms in a sector note — those are single-name valuation questions, one of which is a separate live workstream.



SOURCES & METHOD

Sources & Method (appendix)

Basis. Prices/market caps: Schwab real-time, 2026-08-28. Fundamentals: latest annual as-reported SEC EDGAR XBRL via the house sector-comps engine (15-name curated universe; two Israeli foreign filers noted below), refreshed in the text to each company's most recent dated results release. P/S is market cap over latest fiscal-year revenue — enterprise-value variants were suppressed where EDGAR debt tags did not resolve (nine names), and we substituted plain P/S rather than hand-build EVs from mixed sources. Fiscal calendars differ across the group; where an annual basis is stale (notably Palo Alto, whose FY2026 filing lands with its September 1 report), multiples are re-based in the text to the company's own guidance and labeled.

Threat and spend series. FBI IC3 annual reports (losses/complaints); NIST NVD (CVE counts by NVD publication date, later-rejected records included — counts by CVE-ID year run lower; processing-backlog caveat as stated); CISA Known Exploited Vulnerabilities catalog (v2026.08.27); Microsoft Digital Defense Reports 2024 (password-attack rate) and 2025 (identity-attack growth); Gartner information-security forecasts — July 2025 press release (archived vintage) and the 2Q26 forecast update carrying 2026 at ~\$248.9B (analyst estimates, labeled); CyberArk/Palo Alto Identity Security Landscape 2026 (vendor survey, labeled).

Deals and filings. Completion evidence pulled directly from EDGAR (CyberArk Form 25-NSE and final 6-K of Feb 11, 2026; Splunk Form 25 of Mar 18, 2024) and acquirer completion releases (Google, Mar 11, 2026); consideration terms from the acquirers' announcement releases. Company operating figures from each issuer's own quarterly results release, dated inline.

Regulation. SEC release 33-11216 and the Federal Register; EUR-Lex (regulations 2022/2554, 2024/2847; directive 2022/2555) and European Commission infringement releases; 32 CFR 170 and the DFARS acquisition rule; public laws 119-37/119-75/119-86; the FY2027 President's Budget; NIST's August 2024 PQC standards release; the June 22, 2026 White House post-quantum executive order and OMB M-26-15; NSA CNSA 2.0 documents (flagged below); Cloudflare's published traffic measurements.

Forensic and positioning receipts. Full forensic-accounting engine runs on the three cheapest names and the FINRA short-interest positioning gate on all fourteen US filers are archived with the exhaustive draft, alongside every analysis script and data pull (including the 2026-08-29 one-basis scoreboard refresh script). Positioning summary, stated precisely: over the 90-day window shorts built in only two names — SailPoint (+1.3pp, a new listing with backfilled history, read cautiously) and Gen Digital (+0.3pp, consistent with the forensic flags) — while the heaviest covering came in Tenable (-4.6pp) and Zscaler (-3.1pp); elevated *levels* persist at two-year-percentile highs in Gen Digital, Qualys, and Rapid7.

Disclosed gaps and open items. Check Point files as a foreign private issuer (20-F): its row is hand-built from its own releases with a market-data share count, outside the engine's standard. The scoreboard was refreshed to a single latest-quarter basis on 2026-08-29 — growth in round 1, GAAP margins in round 2 of the claim-verifier process (SentinelOne, Rubrik, Tenable, Qualys, Rapid7 to their July/August 2026 reports; Varonis to its July 28 release; SailPoint to its June 9 release; refresh script archived in the data directory). The CNSA 2.0 dates and both Gartner vintages were corroborated in gating (NSA PDFs still resist automated fetch, so CNSA dates remain paraphrase, never verbatim quotes); the NVD CVE-count basis (publication-date, rejected records included) was independently reproduced. The ~8.5M outage machine count is Microsoft's published estimate. The Microsoft \$20B figure is

presented only as "last disclosed" (January 2023). Steelman gate: PASS (4/5). Claim verifier: two HOLD rounds with corrections applied inline both times, then PUBLISHABLE on the scoped round-3 re-verification (2026-08-29); the September 1 (Palo Alto) and September 3 (Zscaler) fiscal-year reports land after this vintage; both are armed as tripwires (§10) rather than holds on this note, which stands on its stated as-of dates.

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